

Some useful definitions

False Alarm Rate (FAR): defines how frequently a noise event is detected with the same observational properties of the GW candidate

- A. An event is **released publicly** if **$FAR < 2 / \text{day}$**
- B. An event is labeled as **SIGNIFICANT = True** if
 - **$FAR < 1 / 7 \text{ months}$** if found by a **CBC** pipeline
 - **$FAR < 1 / 3 \text{ years}$** if found by a **Burst** pipeline
- C. Otherwise is labeled as **SIGNIFICANT = False**. For non-significant events, no further checks on data quality or parameter estimation is performed, unless an external EM coincidence promotes the significance of the GW candidate (come to my next tutorial on RAVEN !!)

Signal to Noise Ratio (SNR): Identifies how loud is the signal with respect to the typical level of noise. Correlated with the significance of the event, but it is not a statistical measure of how we are confident about its astrophysical origin (glitches in the data can mimic high SNR GW signals, but a correct filtering of the pipelines should downrank them assigning high FAR)

Some useful definitions

Trials factors: Given a single GW real event, multiple pipelines can trigger. Similarly for a noise event. To account for this, the actual rate of false triggers, given a publication threshold, should be corrected by:

$$\text{FAR}_{\text{actual}} = k \times \text{FAR}_{\text{threshold}}$$

A conservative choice for k is the total number of pipelines (+1 given by RAVEN):

- 6 + 1 for CBC
- 2 + 1 for Burst

For instance, for CBC triggers, this choice gives an overall expected rate of triggers of:

- 1 SIGNIFICANT event / month
- 10 non-SIGNIFICANT events / day